

→ Terminology used in Inheritance

i) Base class : class is set of objects which share common behaviour and common attributes. class is not a real-world entity.

ii) Parent class

The superclass name of child class. The class whose features are inherited is known as Parent class.

iii) Child class

The subclass name of child class. The sub class can add its own fields and methods in addition to superclass fields and method.

iv) Reusability

Inheritance supports the method of "reusability" where we create a new class since there is already a class that includes some of code.

Ex

class car extends vehicle {

```
private String name = "Mercedes";  
public static void main (String[] args)  
{
```

```
car my = new car();  
my.horn();  
System.out.println(my.name);  
}
```

variable of a superclass. This type of polymorphism is achieved by method overriding. The base function is said to be overridden.

ex:

```
class Car {
```

```
void run() {
```

```
System.out.println("running");
```

```
}
```

```
class Bike extends Car {
```

```
void run() { System.out.println("Bike"); }
```

```
}  
public static void main (String[] args)
```

```
{  
    Car c = new Car();
```

```
    c.run();
```

```
}
```

```
}
```

Q3. What is Inheritance? Explain with example.

Ans:

Java inheritance is an important pillar of OOP. It will allow you to create flexible and reusable code. The idea behind inheritance in Java is that you can create new class. You can reuse method and fields of the Parent class although you can add new method and fields in your current class also.

Inheritance represents the Parent child relationship.

Syntax :-

```
class Subclass Parent
```

```
{
```

```
}
```

```
class Child extends Parent
```

```
{
```

```
}
```


Q. What is Polymorphism? and its example

Polymorphism :

The word "Polymorphism" means 'having many forms'. Polymorphism is considered one of the important features of OOPS.

Polymorphism allows us to perform a single action in different ways. The word 'Poly' means 'many' and 'morph' means 'forms'. So it is morphisms.

Types of Polymorphism :

i) Compile-time Polymorphism :

The another name of compile time Polymorphism is static Polymorphism.

The method name is same with different parameters and return type.

Overloading of method is called through the reference variable of class.

Compile time Polymorphism is achieved by method overloading and operator overloading.

Method overloading :-

We can have one or more methods with same that give safety by argument numbers types and order.

Method overloading occurs when a class has many methods with the same name with different parameters.

ex

```
void get () { }  
void get (int num) { }  
void get (int num, float num2) { }
```

ii) Run time Polymorphism :

The another name of Run-time Polymorphism is Dynamic method. An overridden method is called through the reference.

Name : Topiya Kevin M.

MCA Sem-I

Sub : OOPS

UID 24MCA016

Date :

Page :



Q 1 Difference between class and objects

class

object

→ class is used as template for declaring and creating object

An object is an instance of class

→ class is a logical entity

Object is a physical entity

→ class is declared once

Object is created many times as per requirement

→ class is declared using class keyword

Object is created through new keyword

class student { }

student s1 = new student { }

ex car { }

ex BMW, Porsche, Maruti